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Formaldehyde Reduction Test ISO 16000-23

1 Sample Information

Sample identification	CALCECRUDA
Batch no.	4400984
Production date	01/2016
Product type	Lime based decorative
Date when sample was received	21/01/2016
Testing (start - end)	28/01/2016 – 12/02/2016

2 Test Method

Method	**Principle**	**Parameter**	**Quantification limit**	**Uncertainty**
ISO 16000-23, EN ISO 16000-3	HPLC-UV	Formaldehyde	3 µg/m^3	22% (RSD) U_m = 2 x RSD= 45 %

Test chamber parameter

Chamber volume, l	119	Temperature, °C	23±1	Relative humidity, %	50±5
Air change rate, 1/h	0.5	Loading ratio, m²/m³	1		

Sample preparation

Product name	**Consumption per layer (g/m^2)**	**Number of layers:**	**Drying time (hours)**
CALCECRUDA	550	2	5

Formaldehyde injection

Formaldehyde was injected into inlet air giving constant concentrations in the chambers during the first 14 days of testing. Average concentration of injected formaldehyde during testing period was 94 µg/m^3.

Deviations from the test method:	None

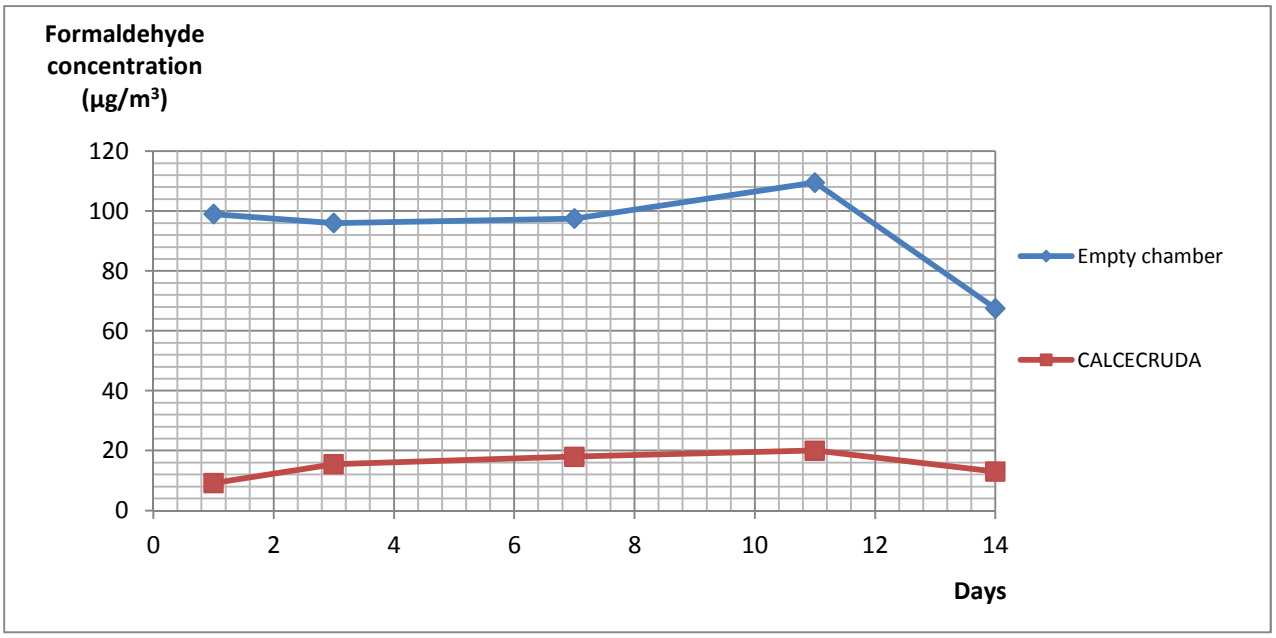
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3 Results

All results are averages of double determinations. An uncertainty of at least ± 22% (includes only chamber testing and analyses; does not include injection) should be considered when evaluating this results.

Sample	Formaldehyde concentration [µg/m³] in chamber					
	Continuous formaldehyde injection					No formaldehyde injection
	1 day	3 days	7 days	11 days	14 days	15 days
Empty chamber Eurofins No. 122	99	96	98	110	68	-
CALCECRUDA	9.2	16	18	20	13	< 3

Test chamber formaldehyde concentration in empty test chamber vs. test chamber with the tested product



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Sample	Formaldehyde consumption [%]				
	1 day	3 days	7 days	11 days	14 days
CALCECRUDA	91	84	82	82	81

Formaldehyde consumption in % was calculated with the following formula:

$$FC = (C_{\text{in}} - C_{\text{out}})/C_{in}$$

Sample	Formaldehyde sorption flux [µg/m^{2}h]				
	1 day	3 days	7 days	11 days	14 days
CALCECRUDA	45	41	40	45	27

Sorption flux was calculated with following formula:

$$F = (C_{\text{in}} - C_{\text{out}})Q_c / A$$

with:
F = sorption flux, µg/m^{2}h
FC = formaldehyde consumption, %
C_{in} = inlet concentration, µg/m^3
C_{out} = chamber concentration, µg/m^3
Q_C = Air flow of chamber = 0.06 m^3/h
A = Area of test specimen = 0.119 m^2

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